MIL-I-82356/23
4 April 1969

MILITARY SPECIFICATION

INSTRUMENTS, SIMULATED, AIRCRAFT TRAINING DEVICES;
MULTI-PURPOSE PRESSURE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The technical requirements for procuring the instrument described herein shall consist of this document and the issue in effect of Specification MIL-I-82356.

1. SCOPE

1.1 This specification covers the detail requirements for simulated multi-purpose pressure indicators for use in aircraft training devices.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-5482	Indicators, Pressure, Synchro, Aircraft 165-Degree Scale
MIL-I-6856	Indicator, Pressure, Synchro 1 1/2 Inch
MIL-I-7085	Indicator, Pressure, Multipurpose, 0-50 PSI
MIL-I-8138	Indicator, Pressure, Synchro, Dual, 0-35 Scale
MIL-I-8391	Indicator, Multi-Purpose Pressure, Autosyn Style, Dual, 0-50 P.S.I., 320 Degree Scale, Type 0-4

FSC 6930

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MIL-S-20708	Synchros, General Specification For
MIL-I-22054	Indicators, Pressure, Synchro, Dual, General Purpose
MIL-G-23337	Gage, Pressure, Dial Indicating
MIL-I-23479	Indicator, Pressure, 1 1/2 Inch, Integrally Lighted
MIL-I-25438	Indicator, Pressure, Type MO-2
MIL-I-25987	Gage, Pressure, Dial Indicating General Purpose. Aircraft
MIL-I-38121	Indicator, Pressure EGU-9/A
MIL-I-82356	Instruments, Simulated, for Aircraft Training Devices, General Specification For

STANDARDS

Military

MS 17996	Indicator, Pressure, 1 1/2 Inch, Integrally Lighted
MS 24512	Gages, Pressure, Aircraft, Multi-Range
MS 25477	Indicators, Pressure, Synchro, Dual General Purpose
MS 27572	Cases, Instruments, Flange-Mounted, Aircraft
MS 28010	Indicators, Pressure - Synchro, Single 2 Inch Size
MS 33639	Case, Instrument, Clamp Mounted, Aircraft

(Copies of specification, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer.)

3. REQUIREMENTS

3.1 General.- The requirements of MIL-I-82356 and this specification shall apply.

3.2 Design basis.- Each simulated aircraft multi-purpose pressure indicator shall conform to the applicable operational instrument specified in Tables I, II, and III of this specification. In each case the activating component shall be coupled to the pointer with a 1:1 ratio.

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3.3 Case.- With the exception of length which shall not exceed the limits specified in the footnotes, cases shall be in accordance with Tables I, II, and III of this specification, as applicable.

3.4 Wiring .- Wiring shall be in accordance with the schematics referenced in Tables I, II, and III as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-I-82356 and as specified in 4.2 through 4.3.

4.2 Tests.- The following tests shall be performed for the instruments referenced in Tables I and II in addition to tests specified in the test requirements of MIL-I-82356.

(a) Backlash of gears

(b) Forward and reverse operation

(c) Running smoothness of gears

(d) No jumping or hesitation of pointers.

4.2.1 Electrical tests.- Electrical tests for instruments referenced in Tables I and II shall consist of checks on electrical zero and running performance. Test equipment shall consist of a transmitter synchro of a size equal to or larger than that employed in the instrument actuator to be tested. The test synchro shall be rotated by a precision dial where zero degrees has been made to correspond to the reference line at the transmitter's true electrical zero in accordance with MIL-S-20708.

4.2.1.1 Electrical zero.- Tests for electrical zero shall be conducted with the test transmitter synchro phase coordinated with the activated receiver synchro in accordance with MIL-S-20708. When the test transmitter synchro dial is at 0° ± 0.5 degrees, the activated synchro receiver pointer (taken respectively in a dual unit) shall read its electrical zero position ± 1.0 degrees.

4.2.1.2 Running tests.- Each receiver synchro, connected in proper phase compatibility as outlined in MIL-S-20708, shall be subjected to the following running tests using the test set-up specified in 4.2.1.

(a) Slow run - Each transmitter synchro shall be rotated clockwise and then counterclockwise at approximately 1 rpm for the full range of the indicator. The receiver synchro indicator pointer shall track smoothly without any discontinuities

(b) Fast run - The test of 4.2.1.2(a) shall be repeated in both directions at a rate of approximately full range in one second. The receiver synchro display pointers shall follow without any discontinuities and not lag the transmitter synchro display pointer by more than 10 degrees.

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4.3 Meter movement tests.- The following tests shall be performed for the instruments in Table III in addition to the tests specified in the synchro test requirements of MIL-I-82356.

4.3.1 Electrical tests.- Electrical tests for instruments referenced in Table III shall consist of checks on electrical zero position scale error tolerance and running performance.

4.3.1.1 Electrical zero and scale error tolerance.- The voltage noted in the tabulation below shall be applied across the electrical terminals of each instrument. The scale error of each indicator shall not exceed the values specified.

VOLTAGE APPLIED ± 0.1 PERCENT	REQUIRED SCALE INDICATION
Volts 0	0 of full scale ± 2 percent
0.2 increasing voltage	0.2
0.4 " "	0.4
0.6 " "	0.6
0.8 " "	0.8
1.0 " "	1.0
0.8 decreasing voltage	0.8
0.6 " "	0.6
0.4 " "	0.4
0.2 " "	0.2
0	0

4.3.1.2 Running performance. Each indicator shall be subjected to the following running tests:

(a) Slow run - Each meter movement shall be deflected both up scale and down scale over the full range of the indicator at a uniform rate of approximately 30 seconds for full scale. The indicator pointer shall track smoothly without any discontinuities

(b) Fast run - Each meter movement shall be deflected both up scale and down scale over the full range of the indicator at a rate of approximately one second for full scale. The indicator pointer shall track smoothly without any discontinuities.

5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery requirements of MIL-I-82356 shall apply.

6. NOTES

6.1 The notes of MIL-I-82356 shall apply.

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CUSTODIANS:
ARMY - AV
NAVY - TD
AIR FORCE - 70

USER ACTIVITIES:
ARMY - AV
NAVY - TD
AIR FORCE - 11, 70

PREPARING ACTIVITY:
NAVY - TD
Project Number 6930-0073

TABLE I - SINGLE SYNCHRO DATA

DESIG-NATION	OPERATIONAL INSTRUMENT BEING SIMULATED	NOMINAL SIZE	*CASE TYPE	SCHEMATIC WIRING PER MIL-I-82356 FIG. NO.
-1S	MIL-I-25438	2 inch	MS 27572	4
-2S	MIL-G-25987		MS 24512	4
-3S	MIL-I-8391	3 inch	MS 27572	4
-4S	MIL-I-7085***	2 inch	MS 28010	4B
-5S	MIL-I-23479	1 1/2 inch	MS 17996	4A
-6S	MIL-I-38121****	2 inch	MIL-I-38121	4
-7S	MIL-I-38121	2 inch	MIL-I-38121	4A
-8S	MIL-I-6856	1 1/2 inch	MS 33639	4

*Maximum case length of simulated instruments may exceed the case length specified in the specifications noted above. However, the maximum case length should not exceed 4 inches without prior approval from the procuring activity.

*** The three position indicator flag for "Water, fuel, and oil" shall be actuated by 26V DC.

**** The instrument shall contain a knob identical in position, appearance, and function as the take off index and counter knob in the operational instrument.

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TABLE II - DUAL SYNCHRO DATA

DESIG-NATION	OPERATIONAL INSTRUMENT BEING SIMULATED	NOMINAL SIZE	*CASE TYPE	SCHEMATIC WIRING PER MIL-I-82356 FIG. NO.
-1D	MIL-I-22054***	2 inch	MS 25477	5A
-2D	MIL-I-22054***	3 inch	MS 25477	5A
-3D	MIL-I-22054***	2 inch	MS 25477	5A**
-4D	MIL-I-22054***	3 inch	MS 25477	5A**
-5D	MIL-I-23479	1 1/2 inch	MS 17996	5A**
-6D	MIL-I-8138	3 inch	MS 33639	5
-7D	MIL-I-5482	2 inch	MS 27572	5
-8D	MIL-I-5482	2 inch	MS 27572	5A**

* Maximum case length of simulated instruments may exceed the case length specified in the specifications noted above. However, the maximum case length should not exceed 4 inches without prior approval from the procuring activity.

** Designation denotes an integrally light instrument.

*** An ON-OFF warning indicator flag shall be provided which will appear identical to the operational instrument and shall be activated by 26V DC with a maximum current consumption of 10 ma.

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TABLE III - SINGLE METER MOVEMENT DATA

DESIG-NATION	OPERATIONAL INSTRUMENT BEING SIMULATED	NOMINAL SIZE	*CASE TYPE	SCHEMATIC WIRING PER MIL-I-82356 FIG. NO.
-1 SM	MIL-I-5482	2 inch	MS 33639	1
-2 SM	MIL-G-23337	3/4 inch	MS 33639	1
-3 SM	MIL-G-23337	1 inch	MS 33639	1
-4 SM	MIL-G-23337	1 1/2 inch	MS 33639	1
-5 SM	MIL-G-23337	2 inch	MS 33639	1

* Maximum case length of simulated instruments may exceed the case length specified in the specifications noted above. However, the maximum case length should not exceed 4 inches without prior approval from the procuring activity.

** An ON-OFF warning indicator flag shall be provided which will appear identical to the operational instrument and shall be activated by 26V DC with a maximum current consumption of 10 ma.

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